Inference often combines conditioning and summarizing

Observed		**Condition**		**Summarize**
y	$\rightarrow$	$p(z \mid y) = \mathcal{N}\left(z \mid \mu_{z\mid y}, \Sigma_{z\mid y}\right)$	$\rightarrow$	$\hat{z} = E[z \mid y] = \mu_{z\mid y}$

Murphy, Book 2 §2.3.2.2–2.3.2.3

$$E[z \mid y] = \int z p(z \mid y)\, \mathrm{d}z$$

- Conditioning preserves the whole posterior distribution and its uncertainty
- Summarizing extracts a quantity from that posterior; for a Gaussian the mean is $\mu_{z\mid y}$

A point estimate needs a loss function

Choose a reconstruction after observing y

$$m = E[z|y], \quad V = \mathrm{Cov}[z|y]$$

Choose any estimate $a \in \mathbb{R}^d$ to reconstruct the unknown vector $z \in \mathbb{R}^d$

$$L(a, z) = \|z - a\|^2, \quad R(a|y) = E[L(a, z)|y]$$

Center the error at the conditional mean

$$r = z - m, \quad z - a = r + (m - a)$$

$$\|z - a\|^2 = \|r\|^2 + 2r^T(m - a) + \|m - a\|^2$$

- A reconstruction is judged by conditional expected squared error
- Given y, the chosen a is fixed; uncertainty remains in z

The conditional mean minimizes squared-error risk

Average the centered expansion Previous slide: $r = z - m$, $V = \mathrm{Cov}[z|y]$

$$E[r|y] = 0$$
$$E[2r^T(m-a)|y] = 2E[r|y]^T(m-a) = 0$$
$$E[\|r\|^2|y] = \sum_{j=1}^{d} V_{jj} = \mathrm{tr}(V)$$

Conditional squared-error risk

$$R(a|y) = \underbrace{\mathrm{tr}(V)}_{\text{Remaining uncertainty}} + \underbrace{\|m-a\|^2}_{\text{Excess risk from the chosen estimate}}$$

Optimal reconstruction

$$a^*(y) = m = E[z|y]$$
$$\min_a R(a|y) = \mathrm{tr}(V)$$

For the conditional Gaussian

$$m = \mu_{z|y}, \quad V = \Sigma_{z|y}$$

- This decomposition needs finite conditional second moments, not Gaussianity.
- The conditional mean minimizes model expected squared error; covariance describes the uncertainty that remains.

Murphy, Book 1 §5.1.5.1; vector decomposition: course elaboration

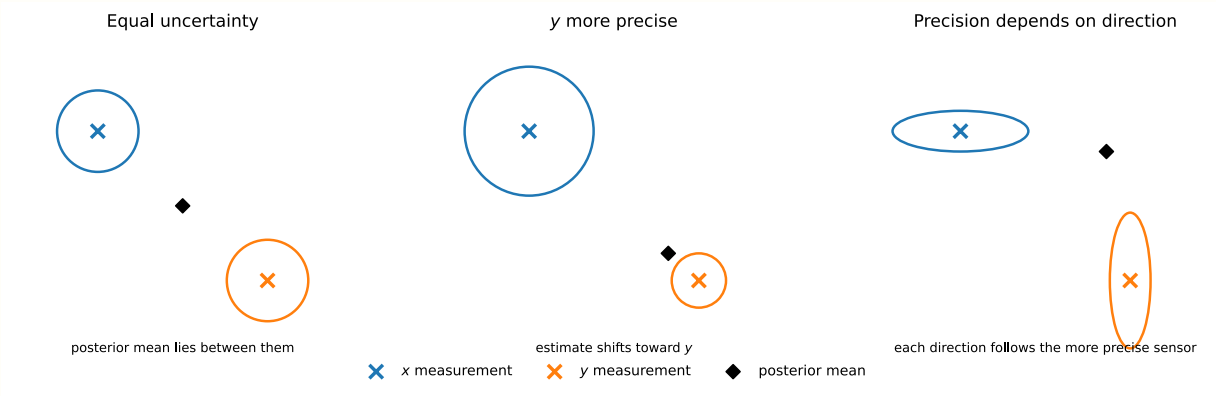
Sensor fusion combines measurements according to their uncertainty

Two noisy measurements of one state

$$z \sim \mathcal{N}(\mu_z, \Sigma_z)\; x \mid z \sim \mathcal{N}(z, \Sigma_x), \quad y \mid z \sim \mathcal{N}(z, \Sigma_y)$$

Same two operations

$$p(z \mid x, y) \quad \rightarrow \quad E[z \mid x, y]$$



Course schematic after Murphy, Book 2 §2.3.2.3, Fig. 2.10; `sensor_fusion_figure.ipynb`

- Equal uncertainty puts the posterior mean between the measurements
- A more precise sensor pulls the posterior mean closer to its measurement
- With anisotropic uncertainty, the weighting can differ by direction